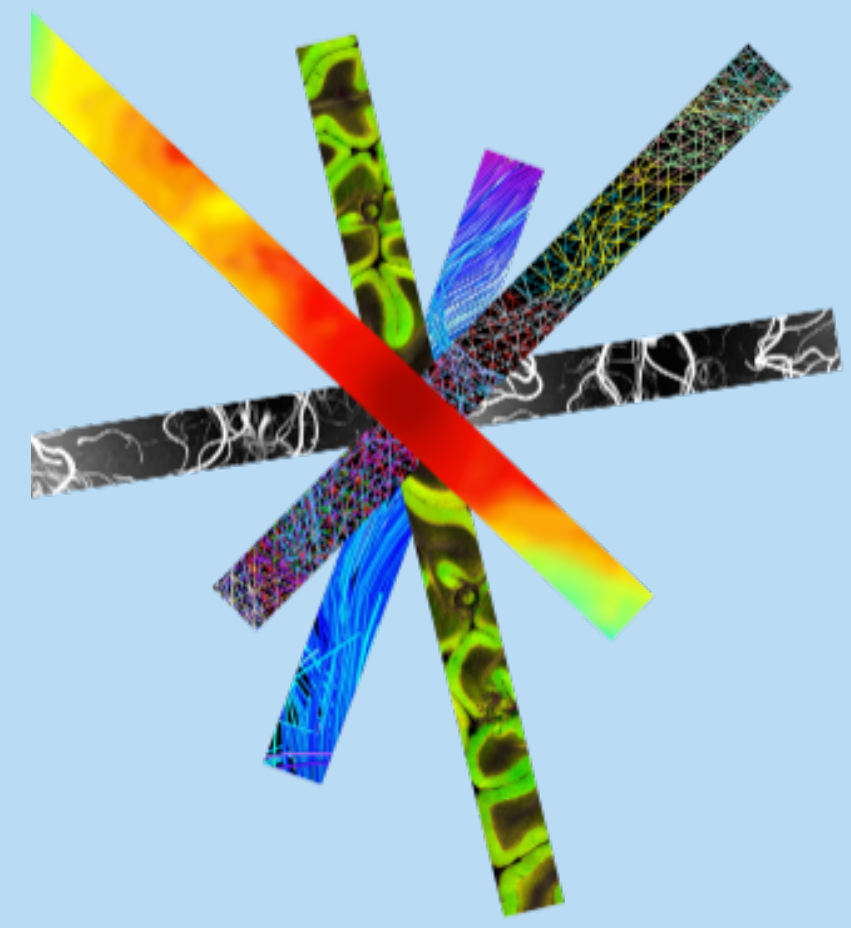




INTRODUCTION

Down syndrome (DS) is the most common genetic cause of intellectual disability and is associated with persistent language and cognitive difficulties, as well as elevated risk for Alzheimer's disease-related neuropathology in adulthood (Hart et al., 2017; Wiseman et al., 2015; Lott and Head, 2019). Expressive language is often disproportionately affected, with weaknesses in morphosyntax, speech intelligibility, naming, and connected speech that meaningfully affect communication and adaptive functioning (Martin et al., 2009).

A converging neurophysiological literature links DS to altered cortical oscillations and network vulnerability. Gamma-band activity supports local cortical computation and cognition (Buzsáki and Wang, 2012), is impaired in DS mouse models (Ruiz-Mejias et al., 2016), and is disrupted early in Alzheimer's disease models, where 40-Hz stimulation has been shown to reduce amyloid burden and improve disease-relevant phenotypes (Iaccarino et al., 2016; Martorell et al., 2019). Human EEG studies in DS similarly report abnormalities across alpha-, beta-, and gamma-range activity (Velikova et al., 2011; Hamburg et al., 2021).

Transcranial photobiomodulation (tPBM) is a noninvasive neuromodulatory approach in which near-infrared light is delivered to the scalp to influence cortical physiology (Hamblin, 2016). Prior studies suggest that tPBM may affect mitochondrial metabolism, cerebral oxygenation, blood flow, electrophysiological activity, and selected cognitive outcomes (Tedford et al., 2015; Vargas et al., 2017; Zomorodi et al., 2019; Saucedo et al., 2021). Preliminary aphasia work also suggests possible language-related benefits and altered functional connectivity (Naeser et al., 2020).

In this pilot randomized, double-blind, sham-controlled trial, adolescents and young adults with DS received 18 sessions of 40-Hz near-infrared tPBM or sham over 6 weeks. This poster focuses on exploratory resting-state fMRI analyses examining whether active tPBM altered functional connectivity in canonical and language-related cortical networks defined using the Glasser360 atlas and Yeo7 network framework (Glasser et al., 2016; Yeo et al., 2011).

METHODS

Image Acquisition

MRI data were acquired on a 3T scanner with multichannel receiver coils using Martinos Center protocols. When feasible, participants completed high-resolution 3D T1-weighted MPRAGE and resting-state fMRI. Prespecified rs-fMRI networks and regions were defined using the Glasser360 atlas and Yeo7 network framework, including language-related cortical regions and canonical Default Mode, Salience, and Central Executive network nodes (Glasser et al., 2016; Yeo et al., 2011).

Preprocessing

rs-fMRI data were preprocessed with longitudinal fMRIPrep v24 and postprocessed in XCP-D using a 36-parameter denoising strategy (Esteban et al., 2019; Mehta et al., 2024). Scans were excluded if >50% of BOLD volumes had framewise displacement >0.5 mm. One participant was excluded because incomplete T1 acquisition precluded adequate preprocessing. Because fMRI data were limited and unevenly available across sessions, rsFC analyses were treated as exploratory.

Statistical Analysis

Functional connectivity values were Fisher z-transformed and averaged within each prespecified network/ROI. Treatment effects were estimated separately using Gaussian mixed-effects models with treatment group as a fixed effect and participant as a random intercept: mean FC z ~ treatment + (1 | participant). Reported effects represent NIR/tPBM minus sham. Because session coverage was sparse, models did not include session or treatment-by-session terms.

Bayesian mixed-effects models (Hadfield, J. D., 2010) used the same structure, diffuse normal fixed-effect priors Normal(0, 1000), and weak inverse-Wishart variance priors (V = 1, nu = 0.002). Models used 100,000 iterations, 3,000 burn-in, and thinning of 5, yielding 19,400 retained posterior samples per model. Posterior summaries included the mean NIR-minus-sham effect and P(NIR > Sham).

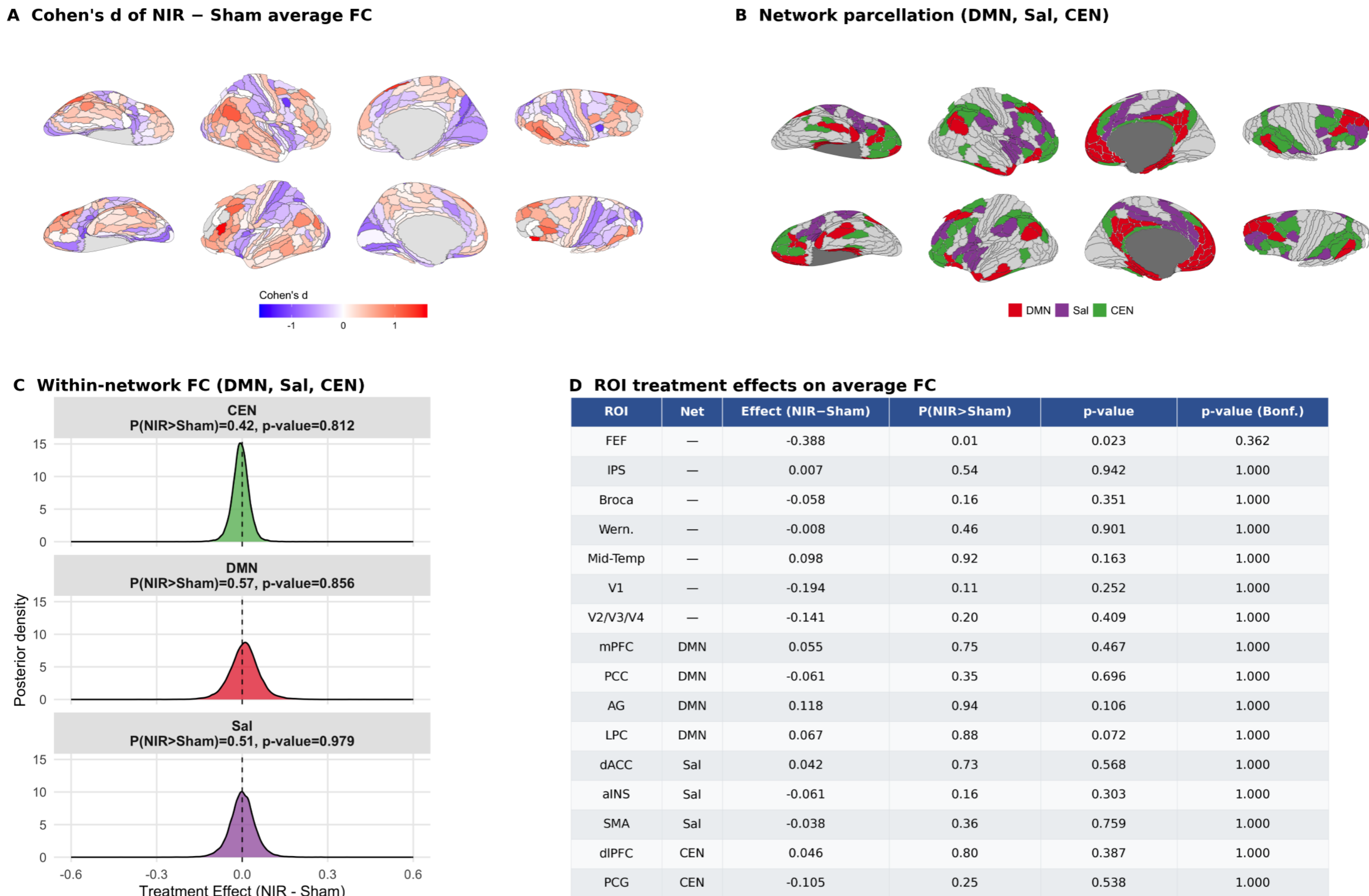


Figure 1. Bayesian analysis of resting-state functional connectivity (rsFC). Fisher z-transformed rsFC was compared between the active tPBM/NIR group (n = 3) and sham group (n = 4). (A) Cohen's d maps for rsFC differences (tPBM - sham). (B) Glasser atlas representation of prespecified Yeo7-derived resting-state networks, including the Default Mode Network (DMN), Salience Network (Sal), and Central Executive Network (CEN). (C) Posterior distributions of treatment effects on within-network rsFC. (D) Region-of-interest treatment effects on average rsFC across prespecified cortical regions, including language-related ROIs. No finding remained significant after correction for multiple comparisons.

Exploratory rs-fMRI analyses did not identify significant treatment effects after correction for multiple comparisons across prespecified regions of interest. At the network level, no robust treatment effects were observed for the Default Mode Network, Salience Network, or Central Executive Network (all uncorrected p-values > .81; posterior P[NIR > Sham] ranged from 0.44 to 0.59). Regional analyses showed modest positive active-treatment effects in the angular gyrus (NIR - Sham = 0.118; P[NIR > Sham] = 0.94; uncorrected p = .106) and mid-temporal region (NIR - Sham = 0.099; P[NIR > Sham] = 0.91; uncorrected p = .163), but these effects were not statistically significant. The only nominal uncorrected regional finding was in the frontal eye fields, where the effect was negative (NIR - Sham = -0.386; P[NIR > Sham] = 0.011; uncorrected p = .023), indicating lower functional connectivity in the active-treatment group relative to sham; this effect did not survive Bonferroni correction (corrected p = .362). Given the very small imaging subsample and exploratory nature of these analyses, rs-fMRI findings should be interpreted cautiously and considered hypothesis-generating.

INCLUSION CRITERIA

Eligibility criteria included age 16–35 years, documented diagnosis of DS, and sufficient spoken English skills to complete the language assessments (i.e., production of at least occasional multiword utterances). Participants were encouraged to attend with a caregiver, to support adherence and to enable caregiver-reported outcomes. Exclusion criteria encompassed seizure disorder, dementia, inability to complete study procedures, primary reliance on non-speech communication or spontaneous speech of fewer than 2-word utterances, recent or anticipated changes in medications or therapies at the investigator's discretion, untreated obstructive sleep apnea, participation in another interfering clinical trial, current pregnancy, active cancer treatment, migraine with aura within the previous 6 months, and conditions increasing risk from NIR exposure. MRI participation was optional; participants with MRI contraindications or unwillingness to undergo MRI remained eligible for all other study procedures.

DISCUSSION & CONCLUSION

Exploratory rs-fMRI analyses did not demonstrate a reliable network-level treatment effect. Modest uncorrected signals in the angular gyrus and mid-temporal cortex were anatomically plausible given their roles in heteromodal association and language-relevant networks, but should be considered hypothesis-generating because of the very small imaging sample and the sensitivity of rsFC estimates to motion (Power et al., 2012). Safety findings were reassuring: all randomized participants completed the trial, no serious adverse events occurred, and no participant withdrew due to adverse events. Thus, repeated 40-Hz NIR tPBM appeared feasible and tolerable in DS, although efficacy remains unproven. This profile is consistent with prior tPBM safety studies reporting no serious adverse events and generally mild, transient side effects (Cassano et al., 2019; Cassano et al., 2022; Coelho et al., 2025). Several limitations temper interpretation. This small pilot was underpowered to detect anything other than large effects, so null findings do not rule out modest benefit. The broad outcome battery increased the risk of both false-negative and false-positive findings, and results from supportive models should be considered provisional. Treatment exposure was not fully uniform because dose titration below 40 J/cm² was permitted and the fifth diode could be omitted for smaller head size. Only one dosing strategy was tested, using a relatively less penetrating wavelength, low average irradiance, targets partly covered by hair, and less-than-daily sessions over 6 weeks. The fixed 40-Hz pulse frequency was not individualized to participant- or region-specific oscillatory abnormalities. Mechanistic sensitivity was also limited by the global gamma metric, optional imaging, very small MRI subgroup, and EEG acquisition across two systems. Finally, the young, dementia-free cohort may have had less manifest Alzheimer-related biology than older DS populations (Lott & Head, 2019).

RESULTS

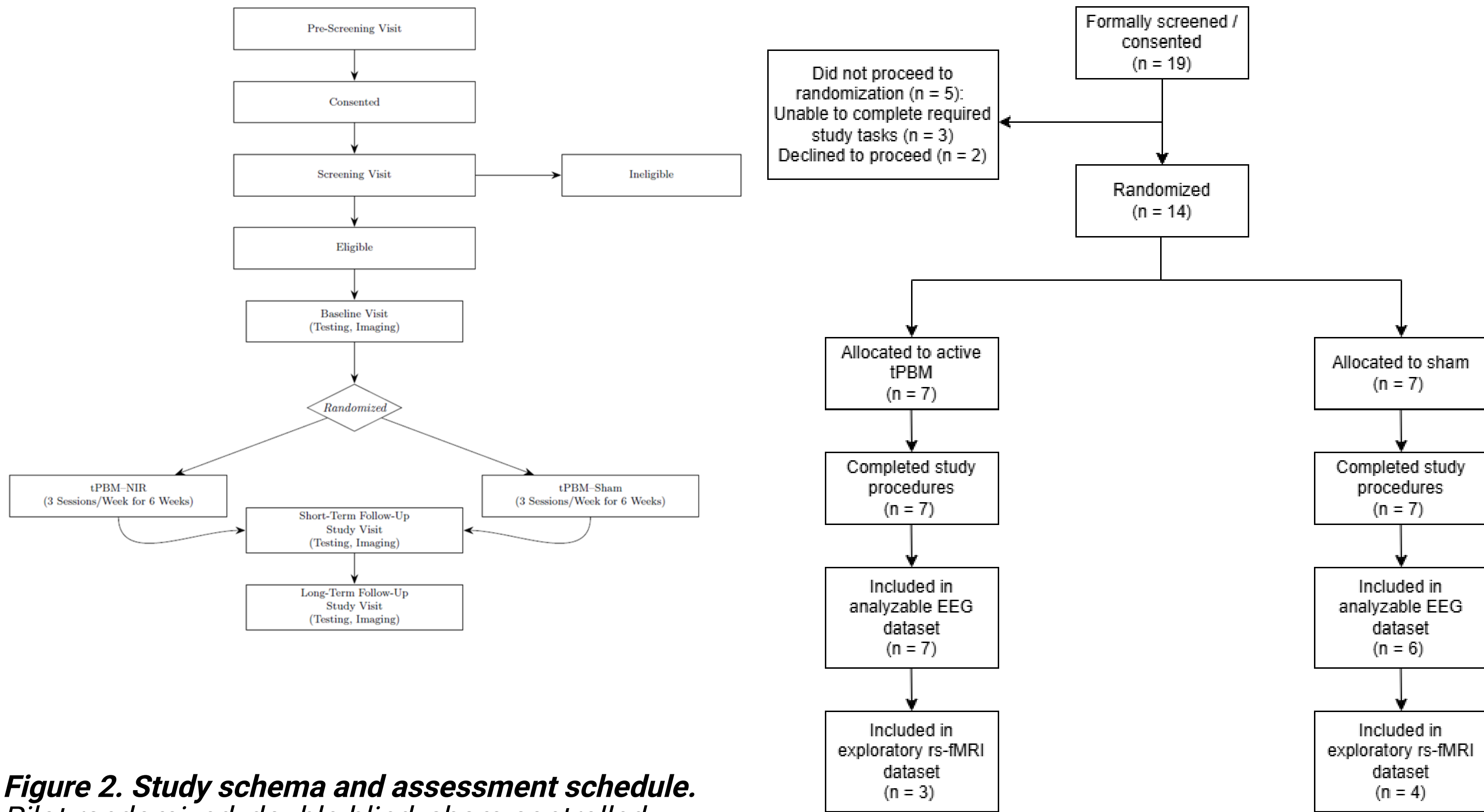


Figure 2. Study schema and assessment schedule. Pilot randomized, double-blind, sham-controlled, parallel-group trial of 40-Hz near-infrared transcranial photobiomodulation (tPBM) in adolescents and young adults with Down syndrome. After screening and baseline assessments, eligible participants were randomized 1:1 to active tPBM or sham and received 18 treatment sessions over 6 weeks (3 sessions/week). Study assessments were performed at baseline, short-term follow-up (within approximately 1 week after the final treatment session), and long-term follow-up (approximately 4 weeks after the final treatment session). Language and cognitive assessments and resting-state EEG were collected longitudinally; resting-state fMRI was optional. Open-label treatment was available per protocol but is not included in the present report.

Figure 3. CONSORT flow diagram and analytic samples. Flow of participants from screening through randomization, study completion, and inclusion in the EEG and exploratory rs-fMRI analyses. All randomized participants completed study procedures. Denominators differ across analytic modalities because rs-fMRI was optional per protocol and some datasets were not available or analyzable after preprocessing/quality-control procedures.

ACKNOWLEDGEMENT

This study was supported by a Private Foundation Grant, MGH Agreement No. **2020A012912**, from the Down Syndrome Research Foundation (United Kingdom). P.V. was supported by a 2025-2026 Research Accelerator Award from the Maud Menten Institute for this study.

REFERENCES

